

Roll No.

TCE-503

B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

GEOTECHNICAL ENGINEERING—II

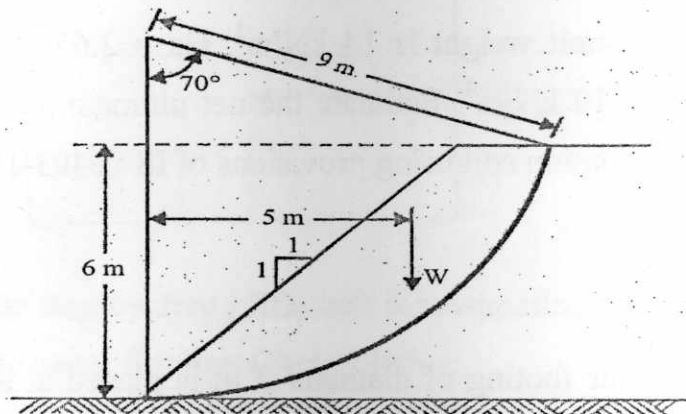
Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a)



The above embankment is made up of cohesive soil with angle of internal friction ($\phi' = 0$) and cohesion ($c = 55$ kPa). The unit weight of the soil is 16 kN/m^3 . Estimate the factor of safety for the trial slip surface shown in the figure. The weight of the sliding mass is 450 kN. Assume there is no tension crack developed in the slope mass. All other geometric details and information are provided in the figure itself.

10 Marks (CO1)

P. T. O.

OR:

- (b) An embankment of 12 m height is inclined at an angle of 38° from the horizontal. Stability analysis using method of slices provided following information :

Σ Normal forces = 950 kN; Σ Shearing Forces = 400 kN and Σ Neutral Forces = 232 kN.

The Length of the failure arc is 29 m. Laboratory tests on the soil provided shear strength parameters $c' = 30$ kPa and $\phi' = 22^\circ$.

10 Marks (CO1)

2. (a) A rectangular footing of size 3 m $\times$ 2 m is placed at a depth of 1.5 m. The Groundwater Table is at the base of the footing. The geotechnical properties of the foundation soil are obtained as $c' = 50$ kPa, $\phi' = 30^\circ$ and dry unit weight is 14 kN/m^3 , $G_s = 2.65$. Assume unit weight of water = 10 kN/m^3 . Estimate the net ultimate load carrying capacity of the foundation soil using provisions of IS : 6403-1981.

10 Marks (CO2)

OR

- (b) A circular footing of diameter 3 m is placed at a depth of 1.0 m. The Groundwater Table is 1 meter above the Ground Surface. The Geotechnical Properties of the foundation soil are obtained as $c' = 80$ kPa, $\phi' = 25^\circ$ and dry unit weight is 15 kN/m^3 , $G_s = 2.65$. Assume unit weight of water = 10 kN/m^3 . Estimate the ultimate load carrying capacity of the foundation soil using provisions of IS : 6403-1981.

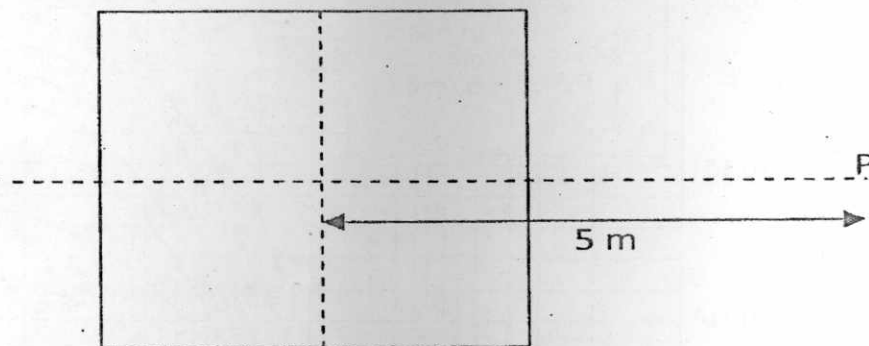
10 Marks (CO2)

3. (a) The size of the rectangular footing is $3 \text{ m} \times 5 \text{ m}$. The stress on the footing is 200 kPa . Estimate the stress at a depth of 4 m from the center point of the footing. Used the Fadum chart in Annexure for the calculation of increase in stress. 10 Marks (CO2)

OR

- (b) The size of the square footing is $3 \text{ m} \times 3 \text{ m}$. The stress on the footing is 250 kPa . Estimate the stress at a depth of 4 m at point P 5 meter away from the center point of the footing as shown in the figure. Use the Fadum chart in Annexure for the calculation of increase in stress.

10 Marks (CO2)



4. (a) Discuss the step by step procedure for estimating the Factor of safety of finite slope using Friction Circle Method. 10 Marks (CO1)

OR

- (b) Discuss the step by step procedure for estimating the factor of Safety of finite slope using Fellinius Method of Slices. 10 Marks (CO1)

5. (a) Write short notes on the following : 10 Marks (CO3)
- (i) Normally consolidated and Overconsolidated Clay
 - (ii) Allowable and Net Allowable bearing pressure

P. T. O.